

CHAPTER 1

INTRODUCTION

1.1. BACKGROUND OF THE STUDY

Effective governance in Local Government Units (LGUs) requires efficient legislative processes that ensure ordinances and resolutions addressing community needs that are created, reviewed, and implemented on time. However, many legislative offices in the Philippines still rely on manual and fragmented systems, which cause delays, inconsistent record-keeping, and difficulty in tracking legislative measures. Additionally, the volume of legislative documents including bills and reports—often overwhelms administrative staff, making it challenging to analyze and extract key information. These limitations reduce governance effectiveness and make it harder to assess whether local laws are working as intended.

The Valenzuela City Sangguniang Panlungsod faces these challenges, particularly with agenda scheduling, legislative tracking, and document management. During interviews with Mr. Jojo Faustino, Administrative Staff of the Sangguniang Panlungsod Office, several problems were identified: the lack of a unified digital system, challenges in managing committee workloads, inefficiencies in scheduling meetings and sessions, and the burden of manually processing legislative documents. These issues highlight the urgent need for a more organized and technology-enabled approach to managing legislative operations.

To address these problems, this capstone project proposes an ERP System with the title, “LGU 2: Legislative Agenda and Calendar Management System with Automated Document Analysis and Summarization Using AI-Classification”. This integrated system will serve as the core scheduling mechanism for the legislative workflow, managing all plenary sessions, committee meetings, public hearings, and stakeholder consultations, while automatically analyzing and classifying legislative documents. By bringing legislative processes and document management together on one digital platform, the system ensures that schedules are followed, deadlines are met, and critical information from documents is automatically extracted and summarized. The system comprises five core modules: Calendar Management to provide centralized oversight of all activities, Session Scheduling to organize when meetings occur, Agenda Creation and Management to structure what gets discussed, Deadline tracking and Monitoring to ensure legal compliance, and a Reminder system to keep officials aware and accountable. These modules create a scalable solution that improves legislative operations, reduces administrative workload, and enhances public service delivery at the municipal level.

1.2. STATEMENT OF THE PROBLEM

1.2.1. MAIN PROBLEM

The legislative operations of the Sangguniang Panlungsod of Valenzuela City are constrained by the continued reliance on manual, paper-based, and fragmented workflows. These traditional practices result in operational inefficiencies, weak coordination among legislative units, limited transparency, and delays in legislative

actions. The lack of a centralized and automated legislative management system hampers effective scheduling, monitoring, documentation, and tracking of ordinances, ultimately affecting the quality of governance and public accountability.

1.2.2. SPECIFIC PROBLEMS

“LGU 2: Legislative Agenda and Calendar Management System with Automated Document Analysis and Summarization Using AI-Classification” aims to answer the following:

1. How does the lack of an automated scheduling system affect the coordination and timeliness of legislative activities?
2. How does the absence of a centralized legislative information platform impact access to accurate and updated legislative data?
3. To what extent do ineffective notification and deadline tracking mechanisms contribute to delays and missed legislative actions?
4. How does manual document processing affect administrative efficiency and accuracy?
5. How does the absence of an integrated ordinance lifecycle tracking system affect transparency and accountability in the legislative process?

1.3. OBJECTIVES OF THE STUDY

1.3.1. GENERAL OBJECTIVE

The primary objective of this capstone project is to design and develop a centralized LGU Legislative Agenda and Calendar Management System with AI-based automated document analysis and classification to enhance the efficiency, coordination, transparency, and organization of legislative scheduling, ordinance workflow management, and document processing within the Valenzuela City Sangguniang Panlungsod.

1.3.2. SPECIFIC OBJECTIVES

“LGU 2: Legislative Agenda and Calendar Management System with Automated Document Analysis and Summarization Using AI-Classification” aims to achieve the following objectives:

1. To develop a centralized platform for managing legislative schedules and agendas to eliminate conflicts and improve coordination.
2. To implement automated deadline tracking and notifications to prevent missed deadlines and procedural delays.
3. To integrate legislative workflow stages with the scheduling system to ensure proper and timely ordinance progression.
4. To develop AI-based document classification and summarization to reduce manual processing and improve information accessibility.
5. To implement conflict detection mechanisms for overlapping schedules and competing legislative activities.

1.4. SIGNIFICANCE OF THE STUDY/PROJECT

1.4.1. CONTRIBUTION TO KNOWLEDGE

This study contributes to the field of Information Technology and E-Governance by introducing a centralized legislative agenda and calendar management system integrated with AI-based automated document analysis and classification, specifically designed for Philippine Local Government Units. Unlike traditional legislative systems that focus solely on document storage or scheduling, this project uniquely combines agenda management, ordinance lifecycle tracking, automated deadline monitoring, intelligent conflict detection, and AI-driven document classification and summarization within a single platform.

1.4.2. BENEFICIARIES OF THE STUDY

“LGU 2: Legislative Agenda and Calendar Management System with Automated Document Analysis and Summarization Using AI-Classification” will benefit the following:

1. **Valenzuela City Sangguniang Panlungsod** - The legislative body benefits from improved efficiency, transparency, and coordination through centralized scheduling, automated document processing, and real-time ordinance tracking.
2. **City Council Members and Committees** - Councilors and committee members gain timely access to accurate schedules, legislative documents, and ordinance statuses, supporting informed decision-making and procedural compliance.

3. **Valenzuela City Constituents** - Citizens benefit from increased transparency, faster legislative action, and improved access to legislative schedules and ordinance progress, fostering public trust and civic engagement.
4. **Future Researchers, Students, and IT Professionals** - The project serves as a case study and reference for system design, AI integration, legislative workflow automation, data security, and e-governance implementation, supporting future academic research and professional innovation.

1.5. SCOPES AND LIMITATIONS

1.5.1. SCOPE

The scope of this study covers the design and development of a Legislative Agenda and Calendar Management System for the Valenzuela City Sangguniang Panlungsod. Specifically, the system includes the following:

- Centralized scheduling of legislative activities such as plenary sessions, committee meetings, public hearings, and consultations
- Integration of legislative workflow stages, including First Reading, Second Reading, and Third Reading of ordinances and resolutions
- Automated deadline tracking and management for legislative timelines and legal requirements

- Multi-level coordination among administrative staff, committee members, and legislative officials
- Automated notification and reminder system for meetings, deadlines, and required actions
- Conflict detection and prevention for overlapping schedules and competing legislative activities
- Public transparency features allowing constituents to view legislative schedules and ordinance progress
- Document organization and AI-based classification and summarization of legislative documents

1.5.2. LIMITATIONS

The study does not include the following functionalities and components:

- Financial management, budgeting, or expenditure tracking related to legislative operations
- Performance analytics, KPI dashboards, or legislative productivity evaluation tools

- Online constituent engagement portals such as public feedback, petitions, or voting mechanisms
- Integration with external national government systems beyond the legislative scope
- Predictive analytics or policy impact forecasting features

1.6. REVIEW OF RELATED LITERATURES

Pernia, R. A. (2023), Pernia (2023) conducted a Philippine-based study using automated text analysis techniques, including keyword extraction and computational text classification, to examine patterns and themes in Philippine political research publications. The study applied Natural Language Processing (NLP) methods to analyze large volumes of political texts, allowing systematic identification of dominant topics and trends without manual reading.

University of the Philippines Center for Integrative and Development Studies (UP-CIDS) (2023), UP-CIDS (2023) documented discussions from a Philippine academic webinar that explored the use of text mining, machine learning, and automated text analysis in political science and public policymaking. The discussions highlighted real-world applications of AI tools for analyzing government documents, policy texts, and institutional records to support governance and administrative efficiency.

University of the Philippines Center for Integrative and Development Studies (UP-CIDS) (2024), The UP-CIDS (2024) concept note discusses the use of

text mining and automated document analysis in policymaking and governance research in the Philippine context. It explains how AI-driven techniques such as text classification, clustering, and summarization can be used to process large sets of policy and government documents efficiently. The paper emphasizes the role of computational tools in improving evidence-based decision-making.

1.7. REVIEW OF RELATED SYSTEMS

Brylkowski & Klikowski (2025), This study uses large language models (LLMs) to analyze legislative texts from the Polish Parliament, conducting automated content analysis on bills and legal proposals. Although not explicitly about calendar management, it shows how AI classification and NLP processing can systematically analyze legislative documents, which is essential for building tools that can help with agenda profiling, topic tagging, and tracking legislative priorities over time. By processing legislative corpora with AI, researchers can identify themes relevant to scheduling and prioritization within a legislative calendar.

Gesnouin et al. (2024), This work demonstrates a state-of-the-art AI model designed specifically to generate concise summaries of legislative proposals. Summarization is a key component of automated document analysis, enabling lawmakers and analysts to digest lengthy legislative texts quickly. In the context of legislative agenda and calendar management, such summarization can help identify priority items, track progress, and align documents with scheduled session activities.

Pressato et al. (2025), This paper presents an automated approach using NLP to organize and analyze amendments in legislative documents. It uses topic clustering and retrieval, which directly aligns with classification of agenda items and legislative texts. This kind of processing supports automated summarization and categorization

— foundational for tools that would assist legislative agenda formation and tracking through the calendar of sessions.

1.8. CONCEPTUAL FRAMEWORK

This conceptual framework illustrates a streamlined Input-Process-Output (IPO) model for a legislative management system, designed to transition from complex data inputs to simplified administrative outcomes. It begins by gathering Input Data—such as schedules, legal deadlines, and ordinance documents—which are then fed into a Processing engine that utilizes AI document classification, automated tracking, and conflict detection to organize and validate the information. These processes generate specific outputs, including conflict-free schedules, transparent public access to documents, and improved stakeholder coordination, all of which ultimately converge to achieve the primary goal of increased efficiency and a reduced administrative workload within the legislative lifecycle.

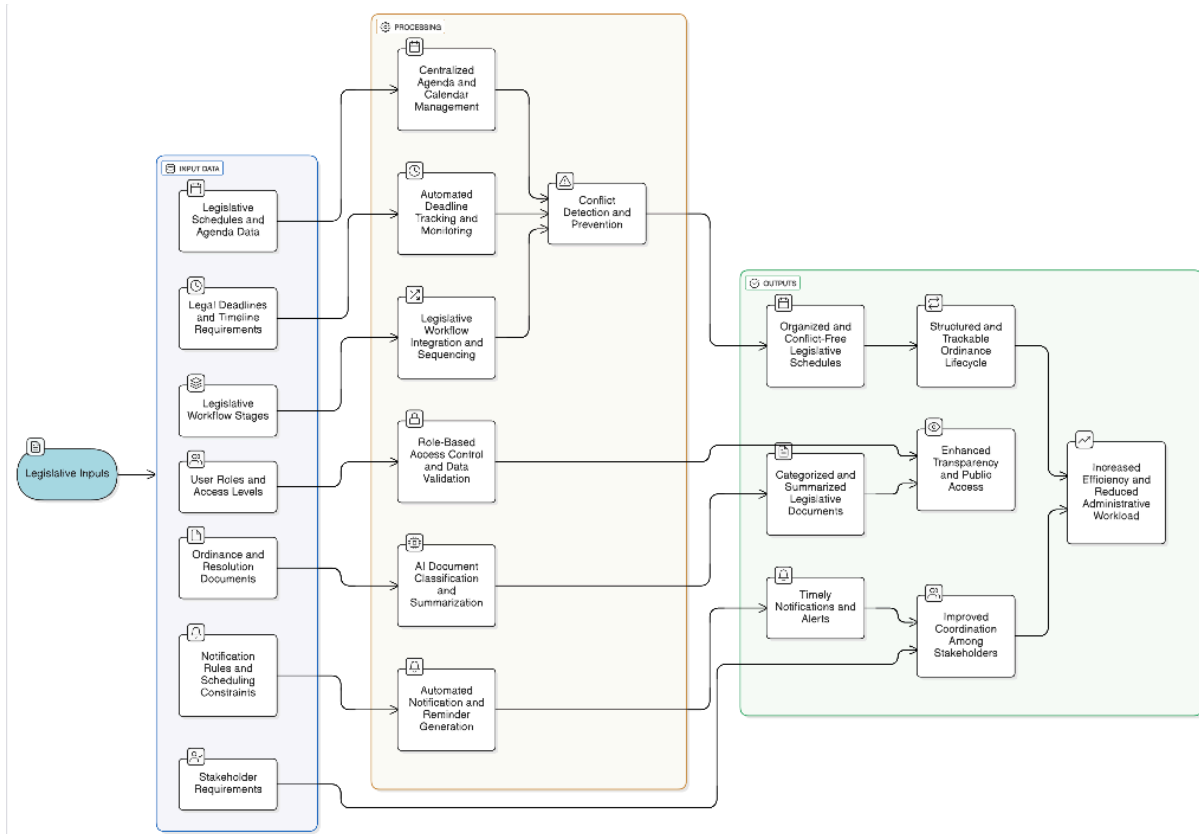


Image 1: IPO Conceptual Framework